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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 03/19/2008 |
| **PI Name:** | McIntyre, Bradley W Ph.D. | | |
| **Project Title:** | Gravity and Integrin Mediated Human Lymphocyte Adhesion | | |
| **Division Name:** | Space Biology | | |
| **Program/Discipline:** | SPACE BIOLOGY | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology | | |
| **Space Biology Cross-Element Discipline:** | (1) Immunology | | |
| **Space Biology Special Category:** | (1) Cell Culture | | |
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| **City:** | Houston | **State:** | TX |
| **Zip Code:** | 77030 | **Congressional District:** | 25 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2000 Space Biology 00-OBPR-01 |
| **Start Date:** | 07/15/2001 | **End Date:** | 07/14/2007 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 3 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 4 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NCE to 7/14/2007, per A. Chu 6/23/06 | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NAG 2-1505 | | |
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| **Performance Goal Text:** | | | |
| | The broad, long term objectives of this proposal are to determine the alterations in normal cellular behavior that are responsible for the defects in lymphocyte function and immunity seen in astronauts returning from space. This proposal will utilize an earth-based Microgravity Analog Culture System (MACS) as a platform to investigate the impact caused randomization of gravity on the basic cellular and biochemical mechanisms of integrin-mediated human lymphocyte adhesion, signaling, and spreading. Our preliminary studies revealed a MACS-induced alteration in integrin a4b1-mediated lymphocyte avidity, spreading, and pseudopodia extension on VCAM-1 and fibronectin. Expression levels of integrin a4b1 was not altered, nor was affinity. Thus, the hypothesis of this proposal is that microgravity alters leukocyte adhesive interactions by influencing the function and not the expression of the integrin a4b1. The specific aims of the proposal are: (1) Confirm and extend the observation that MACS affects the function but not expression of | | |

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| **Task Description:** | the integrin a4b1, (2) Determine whether integrin affinity or cellular avidity is altered by MACS, (3) Define alterations in the integrin adhesion and signaling complex in MACS-treated cells, and (4) Investigate whether the MACS effect can be controlled by pharmacologic or genetic approaches. Quantitative techniques for measuring cell spreading, molecular affinity, and cellular avidity will be employed to compare MACS treated with untreated cells. Fluorescence digital microscopy and conventional biochemistry will be used to analyze the location and signal transduction capacity of integrins after MACS treatment. Cells altered by pharmacologic reagents or transfection of mutant signaling proteins will be examined for abrogation of the MACS induced effects. The identification of defects caused by microgravity will lay the foundation for the development of novel methods for controlling lymphocyte behavior and potentially allowing deleterious effects of microgravity to be eliminated and a normal immune response maintained or restored in astronauts.<br><br>The objective is to determine the lesion caused by microgravity in cellular adhesion and subsequent biological processes such as cell movement and motility.<br><br>It is not clear how microgravity alters the immune response. Our research has pinpointed an alteration in lymphoycte behavior caused by simulated microgravity in a rotating wall vessel. Specficially, cellular adhesion is inhibited. This would cause a fundamental change in cellular movement and recognition during the initiation and maintenance of an immune response. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The proposed research will utilize an Earth-based Microgravity Analog Culture System (MACS) to randomize g and basic cellular and biochemical mechanisms of integrin-mediated lymphocyte adhesion, spreading, and signaling will be examined. The hypothesis of this proposal is that MACS impinges on lymphocyte physiology by influencing the function but not the expression of integrins.<br>Specific Aims:<br><br>Specific aim #1: Confirm and extend the observation that MACS affects the function but not expression of the integrin a4b1.<br><br>Specific aim #2: Determine whether integrin affinity or cellular avidity is altered by MACS.<br><br>Specific aim #3: Define alterations in the integrin adhesion and signaling complex in MACS treated cells.<br><br>Specific aim #4: Investigate whether the MACS effect can be controlled by pharmacologic or genetic approaches. |
| **Task Progress:** | The research utilizes Rotating Wall Vessel bioreactors as an Earth-based system to investigate the impact caused by randomization of gravity on the basic cellular and biochemical mechanisms of integrin-mediated human lymphocyte adhesion, signaling, and spreading. We have found a defect in cellular spreading on the cell adhesion molecule VCAM-1 and the ECM component fibronectin. In soluble ligand-binding assays the affinity of the integrins did not seem to be altered but overall cellular avidity was reduced as determined by quantitative cell adhesion assays utilizing parallel plate flow chambers. Molecular tweezer experiments, which can be used to measure forces at the molecular level, confirm these results. Expression levels of integrin alpha4 beta1 were not altered and activation of PKC could restore normal spreading. The decrease in cellular avidity for extracellular matrix components can explain the original defect observed in cellular spreading. These studies focus our attention on determining the alterations in key intracellular signaling and structural molecules caused by culture in the rotating wall vessel. |
| **Bibliography Type:** | Description: (Last Updated: 03/17/2008) |
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| **Abstracts for Journals and Proceedings** | Andruss BF, Lopez SA, Arya M, Anvari B, McIntyre BW. "Force Measurements with Molecular Tweezers Reveal Reduced Integrin-Mediated Adhesion in Human Lymphocytes Cultured in Simulated Microgravity. " The American Society for Cell Biology, 43rd Annual Meeting, San Francisco, California, December 13-17, 2003.<br>The American Society for Cell Biology, 43rd Annual Meeting, San Francisco, California, December 13-17, 2003. , Dec-2003 |
| **Abstracts for Journals and Proceedings** | McIntyre BW. "Effects of Simulated Microgravity on Integrin Function in Human Lymphocytes." American Society of Cell Biology 43rd Annual Meeting, San Francisco, California, December 13-17, 2003.<br>American Society of Cell Biology 43rd Annual Meeting, San Francisco, California, December 13-17, 2003. , Dec-2003 |
| **Abstracts for Journals and Proceedings** | Woodside DG, Kram RM, Billard MJ, Mitchell JS, McIntyre BW, Vanderslice P. "Domain 4 of vascular cell adhesion molecule-1 (VCAM-1) is a negative regulator of soluble VCAM-1 binding to integrin alpha4 beta1." Annual Meeting of the American Society of Hematology, San Diego, CA, December 6-9, 2003.<br>Annual Meeting of the American Society of Hematology, San Diego, CA, December 6-9, 2003. , Dec-2003 |
| **Abstracts for Journals and Proceedings** | Andruss BF, Lopez SA, Arya M, Anvari B, McIntyre BW. "Force Measurements with Molecular Tweezers Reveal Reduced Integrin-Mediated Adhesion in Human Lymphocytes Cultured in Simulated Microgravity. " 2003 NASA Cell Science Conference, Houston, Texas, February 20-22, 2003.<br>2003 NASA Cell Science Conference, Houston, Texas, February 20-22, 2003. , Feb-2003 |

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| **Articles in Peer-reviewed Journals** | Mitchell JS, Woodside DG, Vanderslice P, McIntyre BW. "Clustering T cell GM1 lipid rafts increases cellular resistance to shear on fibronectin through changes in integrin affinity and cytoskeletal dynamics." submitted, February 2008. , Feb-2008 |
| **Articles in Peer-reviewed Journals** | Billard MJ, McIntyre BW. "CD45RA T-cell activation without proliferation by a partial agonist monoclonal antibody to beta1 integrin." Immunol Cell Biol. 2008 Jan 15; [Epub ahead of print] PMID: 18195724 , Jan-2008 |
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| **Articles in Peer-reviewed Journals** | Woodside DG, Wooten DK, Teague TK, Miyamoto YJ, Caudell EG, Udagawa T, Andruss BF, McIntyre BW. "Control of T lymphocyte morphology by the GTPase Rho." BMC Cell Biol. 2003 Feb 24;4:2. PMID: 12600279 , Feb-2003 |
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